D-Flow, QGF and imperfect critics

Interim research report - methods, completed diagnostics and live studies

Snapshot: 2026-11-05 22:28 MST. Overnight experiments are active. The final morning report is scheduled for 08:00 Phoenix on November 6, 2026. This document does not present unfinished comparisons as results.

Yes: there are 12 D-Flow families. The original eight cover regular/stochastic updates, full/identity Jacobians, and full-flow/Euler endpoint estimates. Four additions test Q averaging and MH. The nearby-action or “slight change to a” idea is the Q-averaging operation; we count it once.

#	Source update	Jacobian	Endpoint	Q score / MH
1	Regular	Full	Full	Raw Q
2	Regular	Full	Euler	Raw Q
3	Regular	Identity	Full	Raw Q
4	Regular	Identity	Euler	Raw Q
5	Stochastic	Full	Full	Raw Q
6	Stochastic	Full	Euler	Raw Q
7	Stochastic	Identity	Full	Raw Q
8	Stochastic	Identity	Euler	Raw Q
9	Regular	Full	Full	Averaged Q
10	Stochastic	Full	Full	Averaged Q
11	Stochastic	Full	Full	Raw Q + MH
12	Stochastic	Full	Full	Averaged Q + MH

Families 1-10 each have separate $\lambda=0$ and $\lambda=1$ conditions. Families 11-12 use $\lambda=T=1$. Thus **12 families produce 22 D-Flow conditions**. QGF and the particle correctors are additional baselines, rather than additional D-Flow families.

What changes in each algorithm?

The source, the action and the critic

The learned BC flow maps an initial Gaussian source z into an action a=F(s,z). Regular D-Flow adjusts z using a critic-derived gradient. It already has randomness from the initial source. Stochastic D-Flow adds fresh movement noise during its source updates.

$$\text{Regular: } z_{new} = z + \eta\,[\kappa\, g(s,z) - \lambda\, z]$$
$$\text{Stochastic: add } \sqrt{2\,\eta\,T}\;\xi, \text{ with } \xi \text{ drawn from } N(0,I).$$

For full/full guidance, g is the gradient of Q(s,F(s,z)) through the full flow. The identity-Jacobian variant replaces the endpoint Jacobian by the identity. The Euler endpoint is an estimate used in guidance; the full-flow endpoint integrates the BC flow.

Parameter	Meaning in this campaign
κ (kappa)	Q-gradient guidance strength; selected separately for each method, task/critic arm, trained model and fixed λ.
η (eta)	Size of each D-Flow source update. QGF instead uses its pseudo-time integration step.
λ (lambda)	Strength of the Gaussian source-prior pull, -λ z. Setting λ=0 removes this term.
T	Stochastic movement temperature. Primary stochastic and MH comparisons fix T=1.
I	Number of D-Flow source updates, swept from 1 through 256.
σ and K	Width of nearby-action perturbations and total antithetic Q queries per average.

Your “slightly change a” idea: evaluate nearby actions

$$\bar{Q}(s,a) = \frac{1}{K} \sum Q_{min}(s, \mathrm{clip}(a + \sigma\, \varepsilon_k, -1, 1))$$

We use the averaged score to guide the update. Stochastic movement noise and nearby-action Q averaging are separate operations, so they can be combined. The same perturbation bank stays fixed throughout one action-generation call and every MH acceptance ratio; it changes at the next environment decision.

MH: propose, check, accept or hold

For λ=T=1, MH uses the target proportional to $\exp[\kappa\, Q_{score}(s,F(s,z)) - \|z\|^2/2]$. It proposes a noisy source move, then checks the target-density change and the forward/reverse proposal probabilities. An accepted proposal replaces z. A rejected proposal keeps the current z and the next iteration continues. This corrects the sampling transition for that specified target; it does not retrain Q or guarantee rapid mixing.

What tasks are running?

1. Native MuJoCo PointMaze navigation

A point robot moves through a maze toward a fixed goal. Its observation is position (x,y), its action has two controls, and the simulator produces an actual next state and reward at every step. We retain learned BC flows and learned IQL Q/V models. Standard task1 runs first, followed by task5 if time permits; these retain the 1000-step benchmark environments.

2. A critic-coverage stress test using native dynamics

The main overnight comparison uses task1 with a goal at (20,20), TRAIN-position starts upstream of a known critic-data hole, and a 128-step budget. The HOLE critic was trained without 10,335 complete transitions for upward actions near (12,4). The FULL critic retained them. Other actions in that region and the full-data BC actor remain available. This isolates a critic-data intervention; it does not make every component unfamiliar with the region.

A declared FULL-QGF-only feasibility rule selected the 128-step budget: 20/24 successes and 24/24 episodes entering the joint state/action hole. No HOLE or new sampler outcome selected that budget. The modified reset distribution and time cap make this a separate stress task.

Fresh standard-task1 baseline health - validation only

Trained seed	BC	Tuned raw QGF	Selected κ
11	0/8	8/8	80
23	0/8	8/8	80
37	0/8	8/8	40

κ=80 is the upper grid edge for two models. These are validation checks, not held-out test performance or evidence that one sampler wins.

Snapshot of completed executions

Study	Validation executions	Held-out executions
Standard maze	1248	0
Coverage stress	1776	0

Partial trajectories are excluded. Repeated executions caused by numerically identical int/float configuration encodings are tracked separately in alias audits; they are not independent new evidence. No incomplete grid is ranked.

Selection and fairness

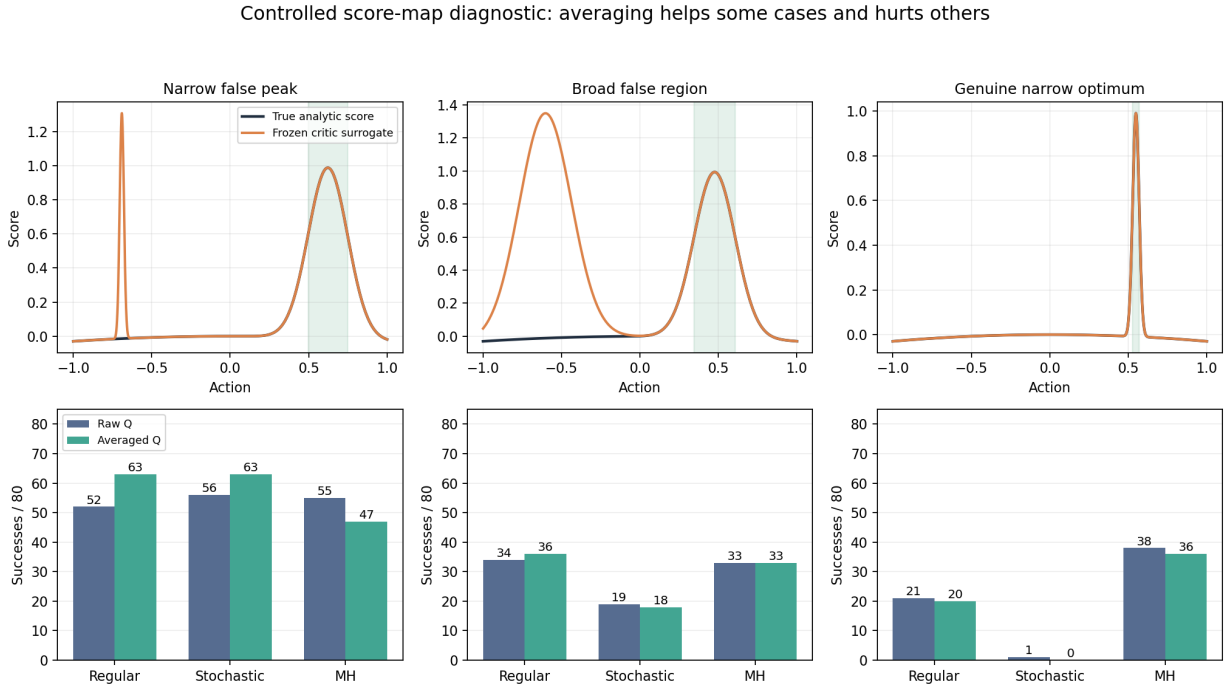
Each method owns its validation-selected κ/η/I and, when applicable, σ/K. The domains include κ=0 and 1, η=.005/.01/.03/.1, I=1/2/4/8/16/32/64/128/256, σ=.01/.03/.1/.2 and K=4/16. Learned-Q prescreening creates bounded simulator-validation shortlists; this can miss good settings. All six FULL/HOLE model selections freeze before any coverage test. The plan uses 20 fresh paired test episodes per selected unique configuration and trained seed. Simulator validation is an oracle tuning resource, and compute is unequal across methods.

Completed diagnostic: narrow and broad Q errors

These are **analytic one-dimensional final-point trials**. They use a known true score and explicitly injected critic artifacts. There is no learned IQL model, robot or next-state dynamics. Twelve cases cover narrow false peaks, broad false regions and genuine narrow optima. All settings were validation-selected before tests.

Method	Narrow false /80	Broad false /80	Real narrow /80
Regular	52	34	21
Stochastic	56	19	1
MH	55	33	38
Avg-Q Regular	63	36	20
Avg-Q Stochastic	63	18	0
Avg-Q MH	47	33	36

Observed: Q averaging helped regular and stochastic methods on narrow false peaks, but reduced MH success there. It did not reliably solve broad errors and could harm a genuine narrow optimum. There is no universal improvement in these cases.



Known score maps and saved held-out counts. Total actual unique evaluations: 1,440 points. Each method has 240 displayed point outcomes across twelve cases; these are not independent trained models.

Raw and averaged-Q methods choose their own settings, so differences combine smoothing and hyperparameter selection. These diagnostics motivate tests on learned critics; they do not establish learned-critic extrapolation accuracy.

Completed diagnostic: wrong target or poor mixing?

This follow-up holds each previously validation-frozen MH target fixed, then compares stochastic updates without MH (ULA) and MH at I=1,4,16,64,256. Numerical integration computes the normalized target's expected success. Fresh random sources are paired across algorithms; no setting is reselected using these outcomes.

Case group / score	Target expectation /80	ULA I256 /80	MH I256 /80
Narrow false / raw	25.27	9	64
Narrow false / averaged	46.41	12	52
Broad false / raw	0.01	13	42
Broad false / averaged	2.27	8	39
Genuine narrow / raw	47.40	1	34
Genuine narrow / averaged	34.67	0	25

Why the broad-error result matters

The raw critic-defined target puts almost all its probability in the wrong region: its expected success is approximately 0.01/80. Yet a finite 256-update MH run achieves 42/80. Its samples have not reached that equilibrium target. Initialization and difficulty moving between modes can temporarily protect successful actions from the misleading critic-defined target.

Implication: a finite MH success rate is insufficient evidence of critic repair or accurate equilibrium sampling. A more accurately sampled wrong target can still produce poor decisions.

What averaging changes

Averaging changes the target itself. It can flatten a narrow false peak, and it can also flatten or shift a genuine narrow optimum. In the genuine-optimum control, the target's expected successes fell from 47.40/80 to 34.67/80. Raw and averaged targets also use their own earlier validation-selected parameters, so this contrast is not an isolated estimate of smoothing alone.

Counts and numerical checks

There are 240 independent case/source random bundles, 960 paired algorithm chains and 4,800 correlated prefix endpoints. The five prefixes are repeated measurements of the same chains. Quadrature grid agreement is 1.07e-5, with tail probability bounded by 2.23e-33. Eight saved partial cases replayed bitwise. An initial quadrature accuracy failure and a later plotting dependency failure remain preserved; finalization used saved arrays only.

These conclusions concern the constructed analytic cases. The native learned-IQL comparison is still running.

How the final report will be organized

Part	What it will contain
Methods	The 12-family table, equations, QGF/IPG/FK definitions and exact distinctions between source noise, nearby-action averaging and MH.
Tasks and data	Native navigation versus analytic points versus previous cube manipulation; full-data BC, FULL/HOLE critic lineage, actual next-state tuples and reset distributions.
Results	Successes/20 for each trained seed, selected parameters, compute, failures/zeros, aliases and partial cells. No missing experiment will appear as a completed zero.
Diagnosis	Coverage-hole exposure, raw/averaged Q, saturation, source movement where available, MH acceptance/mixing and IPG/FK diagnostics.
Reproducibility	Frozen source/checkpoint hashes, grids, seeds, selections, raw trajectories, resume logs and archive integrity; a results bundle and final PDF.

Issues already caught and retained

The new averaged-Q implementation exposed a head-minimum reduction/compiler interaction during preflight. An equivalent elementwise two-head minimum passed finite differences, independent real-network checks and compiled-batch tests before physical release. New kernels use a fixed batch shape for reproducible per-episode actions. The original eight D-Flow implementations and raw QGF remain unchanged.

Integer versus float parameter encoding produced redundant configuration identities and executions. Saved-trajectory alias audits will prevent those records from inflating independent evidence. Failed engineering attempts, sources and logs are retained; no held-out outcome is used to repair a method or select a new setting.

Limits and the next discriminating evidence

Missing conditional training coverage does not automatically imply Q error. A Bellman residual using learned V measures self-consistency, not ground-truth nonterminal Q accuracy. Native tests must show whether a method changes hole exposure and actual success while keeping the critic fixed. Particle, ridge and score approximations and unequal inference/search costs remain explicit limitations.

The standard maze branch may remain partial because its maximum budget exceeds one night. At most two numeric CPU workers run; new expensive starts stop at 07:20, and workers finish or checkpoint by 08:00 Phoenix. Previous frozen reports and archives stay unchanged. No new weak-actor robot sweep has been launched.

Saved source reports for this snapshot

EXPERIMENTS_AND_DIAGNOSIS.txt

controlled_scores/REPORT.txt

matched_target_diagnostic/v2/REPORT.txt

coverage_stress/v2/GATE_REPORT.txt

native_study/DESIGN.json

coverage_stress/comparison/DESIGN.json